

Project3: Interim Data Analysis Plan

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Introduction

The Framingham Heart Study is a landmark prospective cohort study initiated in 1948 with 5,209 participants to investigate the causes and risk factors of cardiovascular disease. Participants have been followed biennially with detailed clinical assessments and surveillance for cardiovascular events. The study has been instrumental in identifying key risk factors and characterizing long-term outcomes such as heart disease and stroke. The dataset used for this project is a subset of the Framingham study and includes laboratory, clinical, questionnaire, and adjudicated event data on 4,434 participants collected across three examination periods from 1956 to 1968.

The primary aim of this analysis is to identify baseline risk factors associated with stroke incidence and to estimate the 10-year probability of stroke across different risk profiles, stratified by sex. This will be accomplished using time-to-event methods, specifically Cox proportional hazards models, to evaluate associations between baseline covariates and the hazard of stroke. Statistically, this involves testing the null hypotheses that individual baseline risk factors are not associated with stroke risk (e.g., $H_0: \beta_j = 0$). A secondary aim is to assess how selected risk factors change across the three examination periods. This component will be descriptive and will evaluate patterns in time-varying covariates, including diabetes status and systolic blood pressure, to assess whether more complex modeling

approaches may be warranted. All analyses will be conducted separately for males and females.

Preliminary Methods

The Framingham dataset is longitudinal, with participants contributing up to three observations across examination periods, resulting in 11,627 total rows. For the primary analysis, the dataset was restricted to baseline measurements from the first examination period, yielding 4,434 unique participants. Participants with prevalent stroke at baseline were excluded ($n = 32$), resulting in a final analytic sample of 4,402 individuals. Because the objective is to evaluate baseline risk factors for incident stroke, only baseline covariates were used. Follow-up time was defined using the time-to-stroke variable (in days), and the analysis was restricted to the first 10 years of follow-up. Participants who experienced a stroke within 10 years were classified as events, while those who did not were censored at 10 years or at their observed follow-up time if shorter. Time was converted from days to years using 365.25 days per year. Baseline covariates of interest included age, systolic blood pressure, diabetes status, use of blood pressure medication, current smoking status, total cholesterol, body mass index (BMI), and history of coronary heart disease. All covariates were measured at baseline and treated as fixed. Categorical variables were coded as binary indicators, and continuous variables were retained on their original scales. Missing data were assessed for all variables. A complete-case analysis was used for modeling, resulting in a final sample size of 4,275 participants. Descriptive analyses were conducted using the full baseline sample prior to exclusion for missingness. A secondary descriptive analysis examined changes in diabetes status and systolic blood pressure across the three examination periods to assess within-individual variability over time. Planned analyses include descriptive summaries of baseline characteristics stratified by sex (Table 1), Kaplan–Meier estimation of stroke incidence over the 10-year follow-up period, and Cox proportional hazards models fit separately

by sex to evaluate associations between baseline risk factors and the hazard of stroke. Variable selection will be performed using a backward selection approach to identify predictors for the final models. Model assumptions, including proportional hazards, will be assessed using standard diagnostic methods.

Table 1: Baseline characteristics overall and stratified by sex.

Variable	Male N=1930	Female N=2472	Overall N=4402
Stroke during follow-up			
No	1881 (97.5%)	2410 (97.5%)	4291 (97.5%)
Yes	49 (2.5%)	62 (2.5%)	111 (2.5%)
Follow-up time (years)	9.40 (1.82)	9.60 (1.54)	9.51 (1.67)
Age	49.73 (8.71)	49.99 (8.63)	49.88 (8.66)
Systolic blood pressure	131.60 (19.40)	133.66 (24.33)	132.76 (22.32)
Blood pressure meds			
No	1868 (97.9%)	2338 (96.1%)	4206 (96.9%)
Yes	40 (2.1%)	96 (3.9%)	136 (3.1%)
Missing, n (%)	22 (1.1%)	38 (1.5%)	60 (1.4%)
Total cholesterol	233.62 (42.39)	239.70 (46.19)	237.01 (44.65)
Missing, n (%)	7 (0.4%)	45 (1.8%)	52 (1.2%)
BMI	26.17 (3.40)	25.57 (4.52)	25.83 (4.08)
Missing, n (%)	4 (0.2%)	13 (0.5%)	17 (0.4%)
Diabetes			
No	1871 (96.9%)	2412 (97.6%)	4283 (97.3%)
Yes	59 (3.1%)	60 (2.4%)	119 (2.7%)
Coronary heart disease			
No	1810 (93.8%)	2405 (97.3%)	4215 (95.8%)
Yes	120 (6.2%)	67 (2.7%)	187 (4.2%)
Current smoker			
No	762 (39.5%)	1471 (59.5%)	2233 (50.7%)
Yes	1168 (60.5%)	1001 (40.5%)	2169 (49.3%)